

Aerosol radiative effect in the European Arctic under high turbidity conditions by sun photometry

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Framework

Aerosols are so closely related with the Arctic radiative budget and it is necessary to understand how the aerosol radiation interaction is modified by aerosol properties.

Database

AERONET AERosol RObotic NETwork (AERONET)
Site: Ny-Ålesund AWI (79°N, 11.9°E)

CIMEL CE318-T (Triple) solar/sky/lunar photometer
Level 2.0 (quality assured data) of AOD and SDA products
Level 1.5 (+ other quality criteria) of inversion products



Detection of aerosol events in Ny-Ålesund

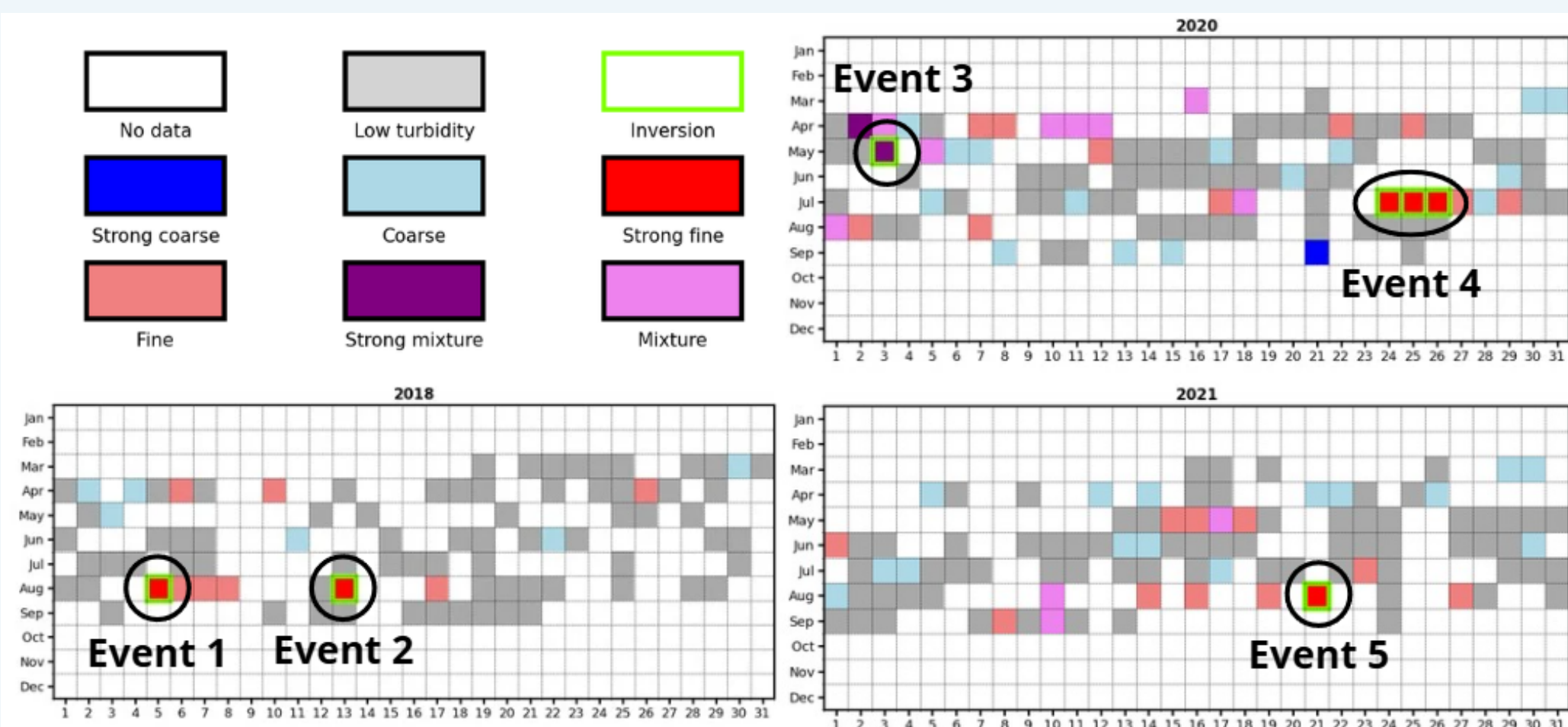
Following the classification method proposed by [Mateos et al. \(2020\)](#), the total aerosol optical depth (AOD) at 500 nm is separated into its fine- and coarse-mode components. These products are obtained from the spectral deconvolution algorithm (SDA) from AERONET.

$$\frac{\text{AOD}_{500}}{\text{Total}} \rightarrow \frac{\text{AOD}_F}{\text{Fine Mode}} + \frac{\text{AOD}_C}{\text{Coarse Mode}}$$

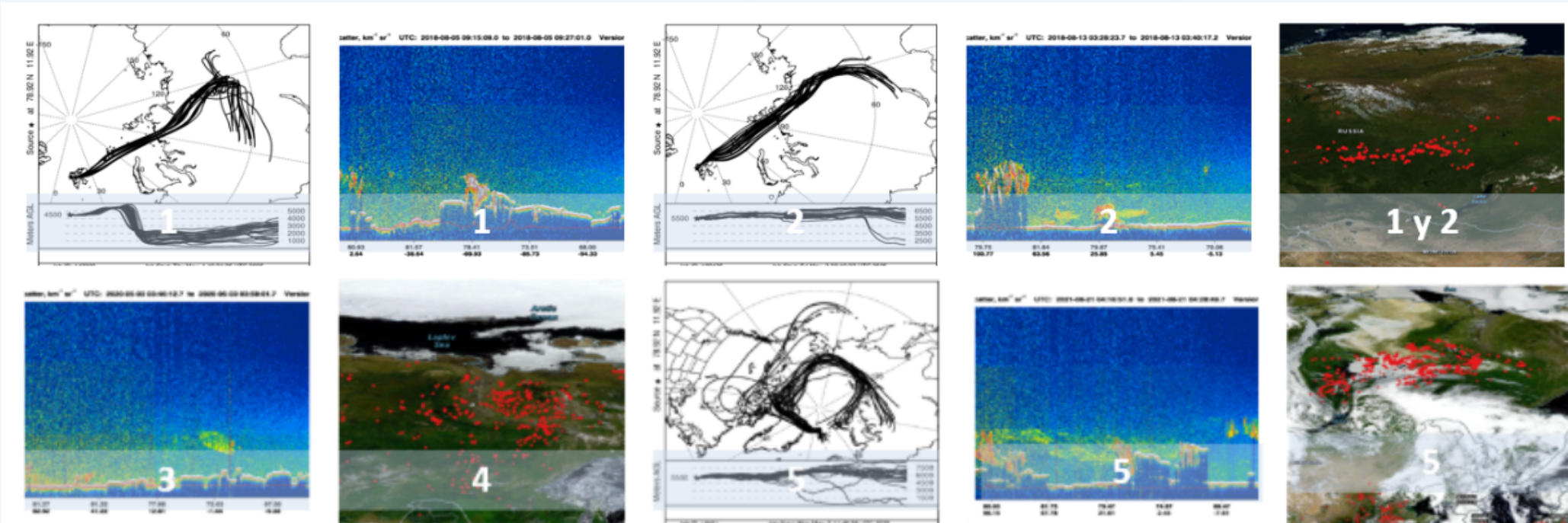
The 85th percentile (P85) of each time series is used as the threshold for identifying aerosol events at Ny-Ålesund:

- **Fine mode:** $\text{P85}(\text{AOD}_F) = 0.078$
- **Coarse mode:** $\text{P85}(\text{AOD}_C) = 0.0208 \rightarrow \text{too low} \rightarrow \text{P93}(\text{AOD}_C) = 0.0352$
- **Mixture mode:** both fine mode and coarse mode criteria must be satisfied.

- ◆ **3 consecutive values** satisfying the corresponding mode criterion.
- ◆ If more than 3 instantaneous $\text{AOD}_{500\text{nm}}$ values are **larger than 0.3**, we assume a moderate-strong event and the event is highlighted by more intense color.
- ◆ Finally, an additional filter based on the aerosol **inversion products availability**.



Using trajectory modeling and remote sensing observations, the origin, vertical distribution, and spatial evolution of the aerosol loads were investigated. This analysis revealed that events 1, 2, 4, and 5 were associated with **wildfire emissions** from the Siberian region of Russia, whereas event 3 was linked to a **dust intrusion** originating from the Gobi and Taklamakan deserts in Mongolia and northwestern China.



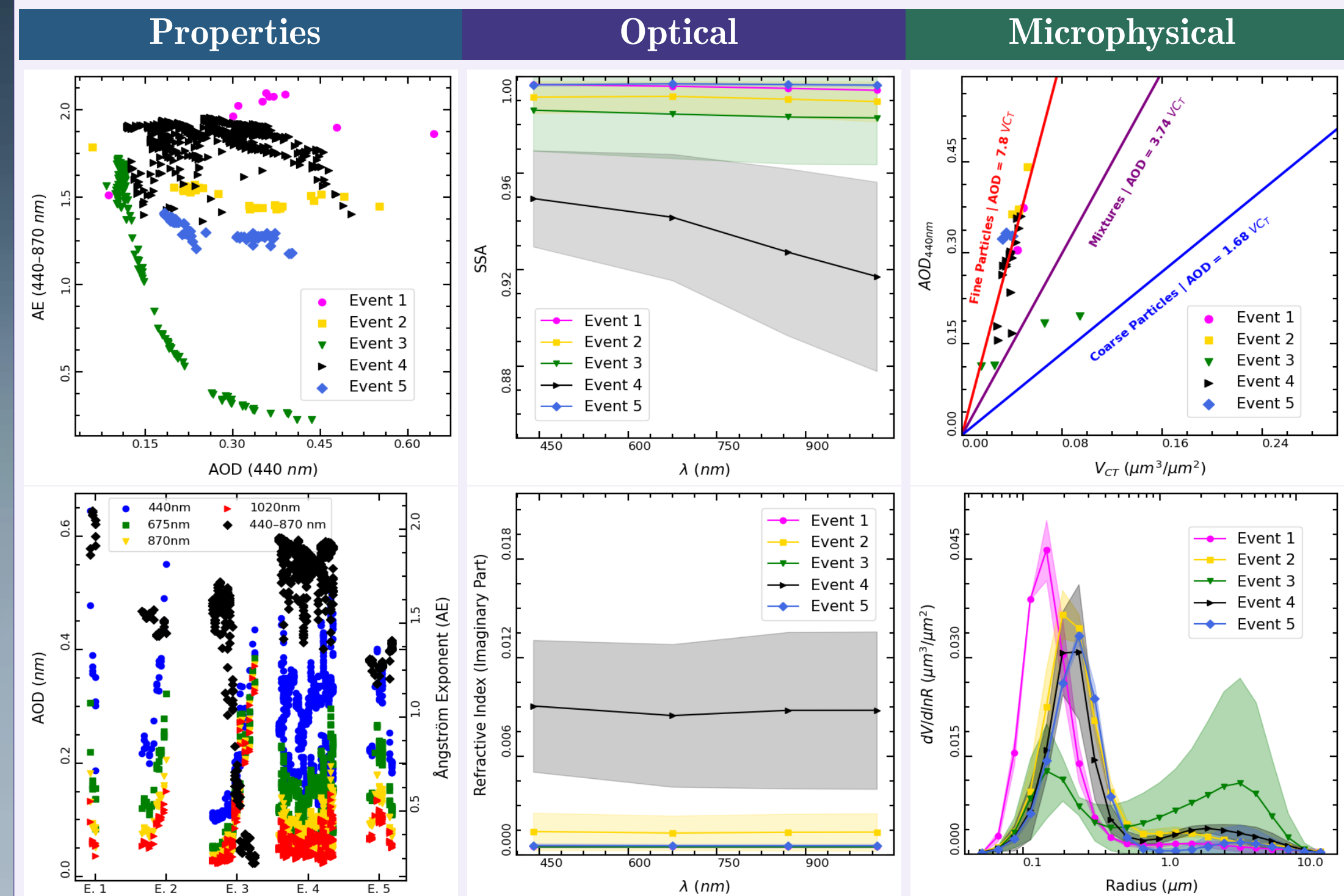
Radiative transfer simulations

libRadtran simulations were performed using aerosol properties from inversion products (α , β , SSA and g) and atmospheric inputs from NASA Giovanni (TOC, PWC and surface albedo).

Simulations were processed in MobaXterm/Linux, obtaining the net radiative flux at the **surface and TOA** for each event in the **UV, VIS and NIR** spectral regions. The net fluxes were summed to obtain the shortwave (SW) flux with and without aerosols, which were then used to calculate the **ARE**.

$$SW_{\text{net}} = UV_{\text{net}} + VIS_{\text{net}} + NIR_{\text{net}} \Rightarrow ARE_{SW} = SW_{\text{net,aerosol}} - SW_{\text{net,aerosol-free}}$$

Some characteristics of the events

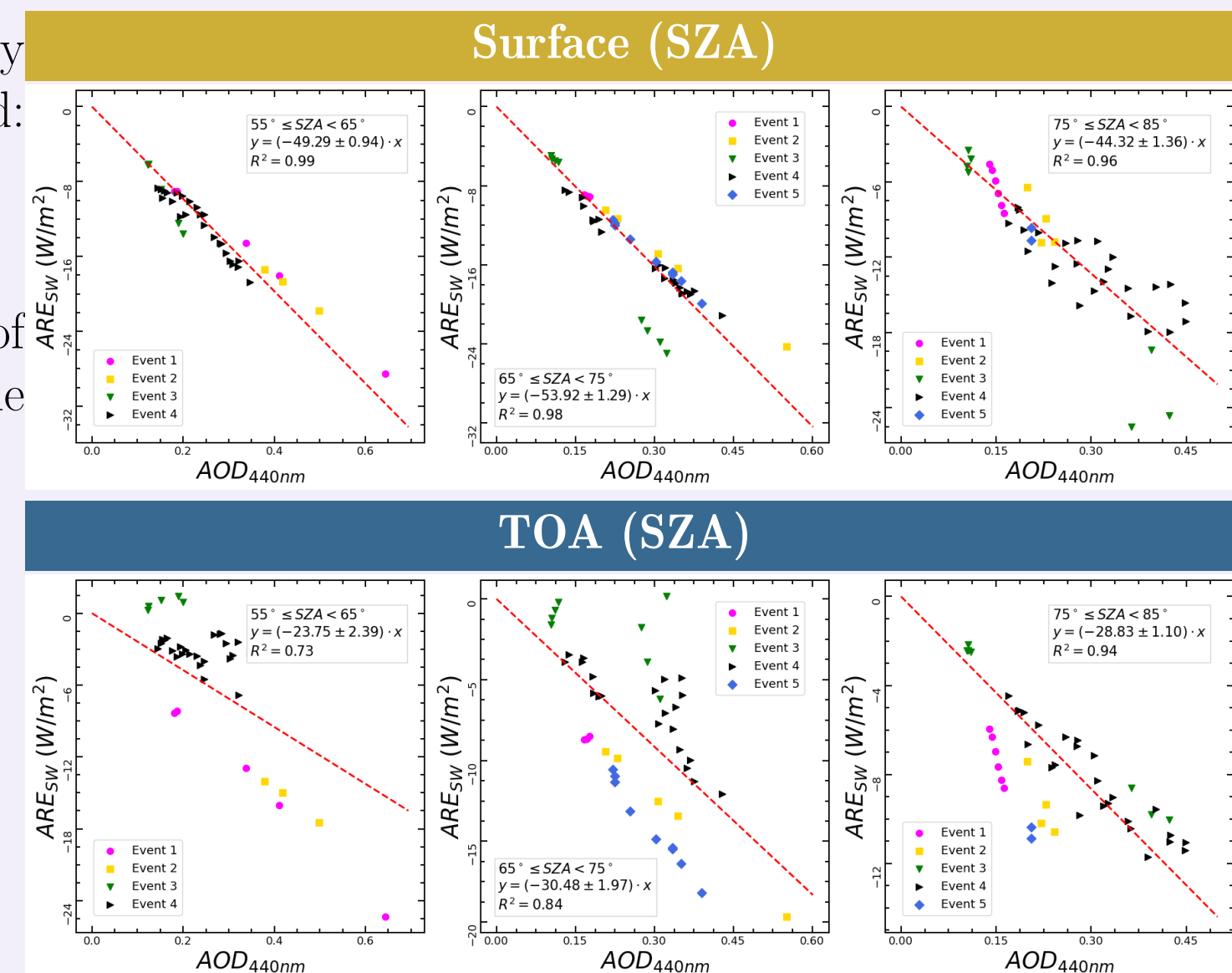
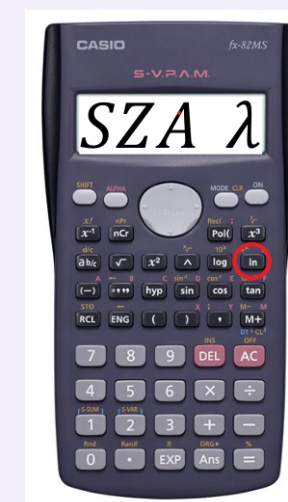


Aerosol radiative estimations

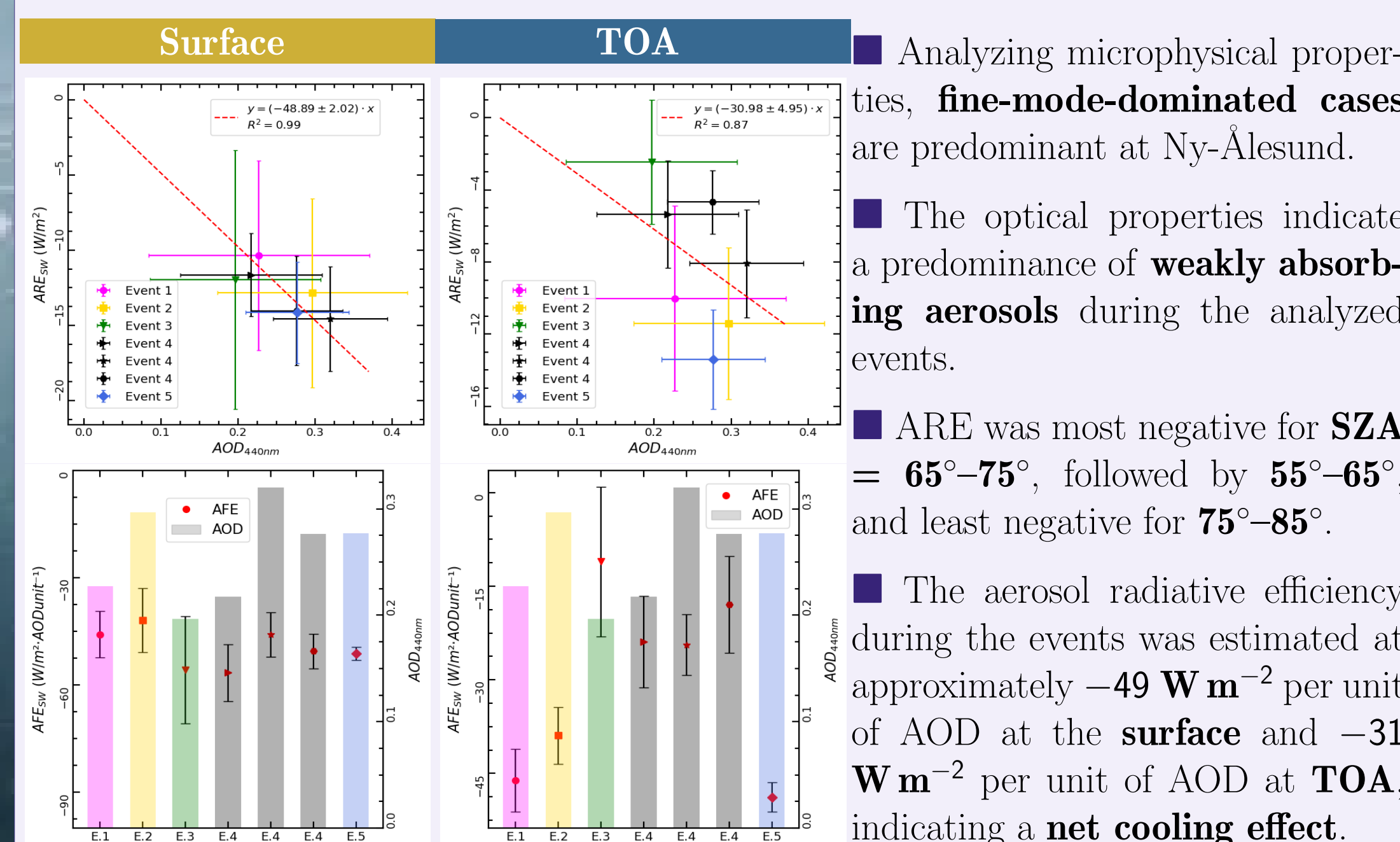
The aerosol forcing efficiency (AFE_{SW}) was also estimated:

$$AFE_{SW} = \frac{dARE_{SW}}{dAOD}$$

As well as the dependence of ARE on the solar zenith angle (SZA).



Main Results



Analyzing microphysical properties, **fine-mode-dominated** cases are predominant at Ny-Ålesund.

The optical properties indicate a predominance of **weakly absorbing aerosols** during the analyzed events.

ARE was most negative for **SZA = 65°–75°**, followed by **55°–65°**, and least negative for **75°–85°**.

The aerosol radiative efficiency during the events was estimated at approximately -49 W m^{-2} per unit of AOD at the **surface** and -31 W m^{-2} per unit of AOD at **TOA**, indicating a **net cooling effect**.

Outlook

To afford daily radiative forcing evaluations and minimize uncertainties in the inversion products, the use of **collocated instruments** and the **GRASP algorithm** should be considered.



AERONET lunar data are still provisional, but they allow the identification of **high-turbidity conditions during the polar night**, although inversion products are not available in this case.

References

Mateos, D. et al. (2020). "Comparison of three different methodologies for the identification of high atmospheric turbidity episodes". In: *Atmospheric Research* 237, p. 104835. ISSN: 0169-8095. [10.1016/j.atmosres.2019.104835](https://doi.org/10.1016/j.atmosres.2019.104835).

